

概率统计 19-20-2(A)标准答案及评分标准

一、选择题

1)D 2) A 3)D, 4) D, 5) A

二、填空题

1) $3/4=0.75$;

2) $2/3$

3) 16

4) 0.8413

5) -0.4

6) 2.8

7) 0.5

8) $\chi^2(10)$

$$9) \quad F(x) = \begin{cases} 0 & x < 1 \\ 0.4 & 1 \leq x < 3 \\ 1 & x \geq 3 \end{cases}$$

$$10) \quad f_Y(y) = \begin{cases} \frac{3-y}{2} & 1 < y < 3 \\ 0 & \text{其它} \end{cases}$$

11) $1/2=0.5$

12) 0.95

13) $\sqrt{17/5} = 1.844$.

三、 (1) $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1; \dots\dots\dots 2'$

$$a \int_0^1 \int_0^{1-y} x(x+y) dx dy = 1; \dots\dots\dots 2'$$

$$a = 8. \dots\dots\dots 1'$$

(2) $f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx. \dots\dots\dots 2'$

$$\text{当 } 0 < y < 1 \text{ 时} \quad f_Y(y) = \int_0^{1-y} ax(x+y) dx = \frac{4}{3}(y^3 - 3y + 2). \dots\dots\dots 2'$$

$$\text{当 } y \leq 0, \text{ 或 } y \geq 1 \text{ 时} \quad f_Y(y) = 0. \dots\dots\dots 1'$$

(3) $f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)} = \begin{cases} \frac{6x(x+y)}{(y^3 - 3y + 2)} & 0 < x < 1-y \\ 0 & \text{其它} \end{cases} \quad (0 < y < 1). \dots\dots\dots 2'$

$$f_{X|Y}(x|0.4) = \begin{cases} \frac{6x(x+0.4)}{(0.4^3 - 3 \times 0.4 + 2)} & 0 < x < 0.6 \\ 0 & \text{其它} \end{cases} = \begin{cases} \frac{1}{0.144} x(x+0.4) & 0 < x < 0.6 \\ 0 & \text{其它} \end{cases} \dots\dots\dots 2'$$

$$P(X < 0.5 | Y = 0.4) = \int_{-\infty}^{0.5} f_{X|Y}(x|0.4) dx = \int_0^{0.5} \frac{1}{0.144} x(x+0.4) dx = 0.6366 \dots\dots\dots 1'$$

四、A1, A2, A3 分别表示 灯管由甲、乙、丙厂家生产；

B 表示抽到的灯管为合格品。则

$$P(A_1) = 0.6; P(A_2) = 0.3; P(A_3) = 0.1; \dots\dots\dots 1'$$

$$P(B|A_1) = 0.95; P(B|A_2) = 0.9; P(B|A_3) = 0.85; \dots\dots\dots 1'$$

(1) $P(B) = P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + P(A_3)P(B|A_3). \dots\dots\dots 2'$

$$= 0.6 \times 0.95 + 0.3 \times 0.9 + 0.1 \times 0.85 = 0.925. \dots\dots\dots 2'$$

(2)

$$P(A_3|B) = \frac{P(A_3B)}{P(B)} = \frac{P(A_3)P(B|A_3)}{P(B)}. \dots\dots\dots 2'$$

$$= \frac{0.1 \times 0.85}{0.925} \approx 0.0919. \dots\dots\dots 2'$$

五、 X 和 Y 的概率密度为：

$$f_X(x) = \begin{cases} e^{-x} & x > 0 \\ 0 & \text{其它} \end{cases}, f_Y(y) = \begin{cases} 2e^{-2y} & y > 0 \\ 0 & \text{其它} \end{cases} \dots\dots\dots 2'$$

X 和 Y 的联合密度为：

$$f(x, y) = \begin{cases} 2e^{-x-2y} & x > 0, y > 0 \\ 0 & \text{其它} \end{cases} \dots\dots\dots 1'$$

$$Z \text{ 的分布函数 } F_Z(z) = P(Z \leq z) = P(X + Y \leq z) \dots\dots\dots 1'$$

当 $z < 0$ 时， $F_Z(z) = 0$;

$$\text{当 } z > 0 \text{ 时， } F_Z(z) = \iint_{x+y \leq z} f(x, y) dx dy \dots\dots\dots 2'$$

$$= \int_0^z \int_0^{z-x} 2e^{-x-2y} dy dx \dots\dots\dots 1'$$

$$= 1 - 2e^{-z} + e^{-2z} \dots\dots\dots 2'$$

Z 的概率密度为

$$f_Z(z) = [F_Z(z)]' = \begin{cases} 2e^{-z} - 2e^{-2z} & z > 0 \\ 0 & z < 0 \end{cases} \dots\dots\dots 1'$$

或者：

X 和 Y 的概率密度为：

$$f_X(x) = \begin{cases} e^{-x} & x > 0 \\ 0 & \text{其它} \end{cases}, f_Y(y) = \begin{cases} 2e^{-2y} & y > 0 \\ 0 & \text{其它} \end{cases} \dots\dots\dots 2'$$

当 $z > 0$ 时

$$f_Z(z) = \int_{-\infty}^{+\infty} f_X(x) f_Y(z-x) dx \dots\dots\dots 2'$$

$$= \int_0^z e^{-x} \times 2e^{-2(z-x)} dx \dots\dots\dots 3'$$

$$= 2e^{-2z}(e^z - 1) \dots\dots\dots 2'$$

$$\text{当 } z \leq 0 \text{ 时， } f_Z(z) = 0 \dots\dots\dots 1'$$

六、设甲厅需要m个座位。设听众选择甲厅的人数为X。

由题意可得： $X \sim b(n, p)$. $n=100; p=0.5$,.....3'

m需要满足：

$$P(X > m) < 2.5\%.....1'$$

$$P(X > m) \approx 1 - \Phi\left(\frac{m - np}{\sqrt{np(1-p)}}\right) = 1 - \Phi\left(\frac{m - 50}{5}\right).....2'$$

$$\text{由 } 1 - \Phi\left(\frac{m - 50}{5}\right) < 0.025; \text{得: } \Phi\left(\frac{m - 50}{5}\right) > 0.975$$

$$\frac{m - 50}{5} > 1.96.....2'$$

$m > 59.8$,故甲厅至少需要准备60个座位才能满足要求。...1'

$$\text{七、(1) 似然函数为: } L(\theta) = \prod_{i=1}^n f(X_i, \theta) = \prod_{i=1}^n 2e^{-2(X_i - \theta)} = 2^n e^{-2 \sum_{i=1}^n X_i} e^{2n\theta}2'$$

$$\theta \leq X_1, \dots, \theta \leq X_n; \text{即: } \theta \leq \min(X_1, \dots, X_n).....1'$$

显然 $L(\theta)$ 是 θ 的单调增函数, 且 $\theta \leq \min(X_1, \dots, X_n)$,

所以当 $\theta = \min(X_1, \dots, X_n)$ 时, $L(\theta)$ 取得最大值.....2'

根据最大似然估计的定义, $\hat{\theta} = \min(X_1, \dots, X_n)$ 是参数 θ 的最大似然估计。...1'

$$(2) E\hat{\theta} = E \min(X_1, \dots, X_n).....1'$$

$$\text{因为 } X_i \geq \theta, \text{所以 } \min(X_1, \dots, X_n) \geq \theta; \text{故 } E\hat{\theta} = E \min(X_1, \dots, X_n) \geq \theta.....2'$$

$$\text{所以 } \hat{\theta} \text{不是 } \theta \text{的无偏估计。}.....1'$$

总体的分布函数;

$$F_X(x) = \begin{cases} 1 - e^{-2(x-\theta)} & x \geq \theta \\ 0 & x < \theta \end{cases}$$

记 $Z = \hat{\theta} = \min(X_1, \dots, X_n)$ 的密度函数

$$f_Z(z) = n[1 - F_X(z)]^{n-1} f(z)$$

$$= \begin{cases} 2ne^{-2n(z-\theta)} & z \geq \theta \\ 0 & z < \theta \end{cases}$$

$$E\hat{\theta} = EZ = \int_{\theta}^{\infty} z 2ne^{-2n(z-\theta)} dz$$

$$= \theta + \frac{1}{2n} \neq \theta, \text{所以, } \hat{\theta} = \min(X_1, \dots, X_n) \text{不是 } \theta \text{的无偏估计}$$

八、(1) $n = 25, \alpha = 0.05,$

检验统计量 $T = \frac{\bar{X} - 16}{S_n} \sqrt{n} \mid H_0 \sim t(n-1)$

拒绝域: $D = \{|T| > t_{\alpha/2}(n-1)\} = \{|T| > 2.064\} \dots\dots\dots 5'$

$\bar{x} = 15, s_n = 3$

T 的观测值: $T = \frac{15-16}{3} \sqrt{25} = -5/3$

$|-5/3| < 2.064,$

所以, 不能拒绝原假设 $\dots\dots\dots 1'$

(2) σ^2 的置信度为95%的置信区间为: $[\frac{(n-1)S_n^2}{\chi_{0.025}^2(24)}, \frac{(n-1)S_n^2}{\chi_{0.975}^2(24)}] \dots\dots 2'$

$= [\frac{24 \times 3^2}{39.36}, \frac{24 \times 3^2}{12.4}] = [5.49, 17.42] \dots\dots\dots 2'$